

On Fuglede's conjecture in finite Abelian groups

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Spectral set

Definition

A bounded measurable subset $\Omega \subseteq \mathbb{R}^n$ with $\mu(\Omega) > 0$ is called spectral, if there is a subset $\Lambda \subseteq \mathbb{R}^n$ such that the set of exponential functions $\{e_\lambda(x)\}_{\lambda \in \Lambda}$ is a complete orthogonal basis, where $e_\lambda(x) = e^{2\pi i \langle \lambda, x \rangle}$, that is

$$\langle e_\lambda, e_{\lambda'} \rangle = \int_{\Omega} e_{\lambda - \lambda'}(x) dx = \delta_{\lambda, \lambda'} \mu(\Omega),$$

where $\delta_{\lambda, \lambda'} = 1$ if $\lambda = \lambda'$ and 0 otherwise, and every $f \in L^2(\Omega)$ can be expressed as

$$f(x) = \sum_{\lambda \in \Lambda} a_\lambda e_\lambda(x),$$

for some $a_\lambda \in \mathbb{C}$. Λ is called the spectrum of Ω , and (Ω, Λ) is called a spectral pair in $\mathbb{R}^n$.

Tiling set

Definition

A subset $A \subseteq \mathbb{R}^n$ tiles $\mathbb{R}^n$ by translation, if there is a translation set $T \subseteq \mathbb{R}^n$ such that almost all elements of $\mathbb{R}^n$ have a unique representation as a sum $a + t$, where $a \in A$, $t \in T$. We will denote this by

$$A \oplus T = \mathbb{R}^n.$$

T is called the tiling complement of A , and (A, T) is called a tiling pair in $\mathbb{R}^n$.

Fuglede's conjecture

Conjecture (Fuglede's conjecture, 1974)

A subset $\Omega \subseteq \mathbb{R}^n$ of positive and finite Lebesgue measure is a spectral set if and only if it tiles $\mathbb{R}^n$ by translation.

Remark

Fuglede's conjecture holds when the tiling set or the spectrum is a lattice subset of $\mathbb{R}^n$. (Fuglede, 1974)

Fuglede's conjecture false in $\mathbb{R}^n$, $n \geq 3$

- $S \nrightarrow T$ in $\mathbb{R}^n$, $n \geq 5$. (Tao, Math. Res. Lett., 2004.)
- $T \nrightarrow S$ in $\mathbb{R}^n$, $n \geq 5$. (Kolountzakis and Matolcsi, Forum Math., 2006.)
- ...
- $S \nrightarrow T$ in $\mathbb{R}^n$, $n \geq 3$. (Kolountzakis and Matolcsi, Collect. Math., 2006.)
- $T \nrightarrow S$ in $\mathbb{R}^n$, $n \geq 3$. (Farkas, Matolcsi and Móra, J. Fourier Anal. Appl., 2006)

Remark

All of these counterexamples are first constructed in finite abelian groups, and then lift them to vector spaces over real field.

Direction 1: Under additional conditions

- Fuglede's conjecture holds for convex sets in $\mathbb{R}^2$. (Iosevich, Katz and Tao, Math. Res. Lett., 2003.)
- Fuglede's conjecture holds for convex sets in $\mathbb{R}^3$. (Greenfeld and Lev, Anal. PDE, 2017.)
- Fuglede's conjecture holds for convex sets in $\mathbb{R}^n$. (Lev and Matolcsi, Acta Math., 2022.)

Direction 2

Question

For which abelian group G , Fuglede's conjecture holds?

Remark

Fuglede's conjecture holds in $\mathbb{Q}_p$, the field of p -adic numbers. (Fan, Fan, Liao and Shi, Math. Ann., 2019.)

Finite abelian groups

Theorem (The Fundamental Theorem of Finite Abelian Groups)

Let G be a finite abelian group, then $G \cong \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_t}$, where t is the number of generators, $\mathbb{Z}_{m_i}$ ($i = 1, \dots, t$) is a cyclic group of order m_i , and $m_1 \mid m_2 \mid \cdots \mid m_t$, $m_1 > 1$.

Preliminaries

- $G \cong \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_t},$
- $g = (g_1, \dots, g_t) \in G,$ where $g_i \in \mathbb{Z}_{m_i},$
- Inner product: $\langle g, h \rangle := \sum_{i=1}^t g_i h_i \frac{m_t}{m_i},$ for any $g, h \in G,$
- Characters: $\chi_g(h) = e^{\frac{2\pi i \cdot \langle g, h \rangle}{m_t}}.$

Group ring

- $\mathbb{Z}[G]$: the group ring of G over $\mathbb{Z}$,
- $A \in \mathbb{Z}[G]$, then $A = \sum_{g \in G} a_g g$, where $a_g \in \mathbb{Z}$,
- for any $\chi \in \hat{G}$, define $\chi(A) = \sum_{g \in G} a_g \chi(g)$.

Spectral set

Definition

Let G be a finite abelian group. A subset $A \subseteq G$ is said to be spectral if there is a subset $B \subseteq G$ such that

$$\{\chi_b : b \in B\}$$

forms an orthogonal basis in $L^2(A)$, the vector space of complex valued functions on A with Hermitian inner product $\langle f, g \rangle = \sum_{a \in A} f(a) \overline{g(a)}$. In such case, the set B is called a spectrum of A , and (A, B) is called a spectral pair.

Remark

The pair (A, B) being a spectral pair is equivalent to

$$|A| = |B| \text{ and } \chi_{b-b'}(A) = 0 \text{ for all } b \neq b' \in B.$$

Remark

Let $M = \frac{1}{\sqrt{|A|}}(\chi_b(a))_{a \in A, b \in B}$, then M is a Hadamard matrix.

Tiling set

Definition

Let G be a finite abelian group. A subset $A \subseteq G$ is said to be a tile if there is a subset $T \subseteq G$ such that each element $g \in G$ can be expressed uniquely in the form

$$g = a + t, \quad a \in A, \quad t \in T.$$

We will denote this by

$$G = A \oplus T.$$

The set T is called a tiling complement of A , and (A, T) is called a tiling pair.

Finite abelian groups with 2 generators

Fuglede's conjecture holds in

- $\mathbb{Z}_p^2$. (Iosevich, Mayeli and Pakianathan, Anal. PDE, 2017.)
- $\mathbb{Z}_p \times \mathbb{Z}_{p^2}$. (Shi, J. Lond. Math. Soc. (2), 2020.)
- $\mathbb{Z}_p \times \mathbb{Z}_{p^n}$. (Z., SIDMA, 2023.)
- $\mathbb{Z}_p \times \mathbb{Z}_{pq}$. (Kiss and Somlai, Proc. Amer. Math. Soc., 2021.)
- $\mathbb{Z}_{pq} \times \mathbb{Z}_{pq}$. (Fallon, Kiss and Somlai, J. Funct. Anal., 2022.)

Two important properties of sets in $\mathbb{Z}_p \times \mathbb{Z}_{p^n}$

Define

$$H(u, t) := \{x \in \mathbb{Z}_p \times \mathbb{Z}_{p^n} : \langle x, u \rangle = t\},$$

and

$$H_A(u, t) := H(u, t) \cap A,$$

for $u \in \mathbb{Z}_p \times \mathbb{Z}_{p^n}$, $t \in \mathbb{Z}_{p^n}$ and $A \subseteq \mathbb{Z}_p \times \mathbb{Z}_{p^n}$.

Lemma (Equi-distributed property)

Let $A \subseteq \mathbb{Z}_p \times \mathbb{Z}_{p^n}$ and $u \in \mathbb{Z}_p \times \mathbb{Z}_{p^n}$. Then $u \in \mathcal{Z}_A$ if and only if $|H_A(u, t)| = |H_A(u, t')|$ for all $t' \equiv t \pmod{p^{n-1}}$.

Lemma

Let p be a prime and set $\zeta = \zeta_{p^e}$, a primitive p^e -th root of unity. Let $c = \sum_{i=0}^{p^e-1} c_i \zeta^i$, where $c_i \in \mathbb{Z}$. Then $c = 0$ if and only if $c_i \equiv c_j$ for any i, j with $i \equiv j \pmod{p^{e-1}}$.

Two important properties of sets in $\mathbb{Z}_p \times \mathbb{Z}_{p^n}$

Lemma (Divisibility property)

Let $A \subseteq \mathbb{Z}_p \times \mathbb{Z}_{p^n}$. If $(a, p^{i_1}), (0, p^{i_2}), \dots, (0, p^{i_s}) \in \mathcal{Z}_A$ for some $a \in \mathbb{Z}_p$ and $0 \leq i_1 < i_2 < \dots < i_s \leq n-1$, then $p^s \mid |A|$.

Lemma (Malikiosis and Kolountzakis, Discrete Anal., 2017.)

Let $A \subseteq \mathbb{Z}_N$ and $p^m \parallel N$. If $\frac{N}{p^{i_1}}, \frac{N}{p^{i_2}}, \dots, \frac{N}{p^{i_s}} \in \mathcal{Z}_A$ for some $0 < i_1 < \dots < i_s \leq m$, then $p^s \mid |A|$.

Tile $\Rightarrow$ Spectral

If (A, T) is a nontrivial tiling pair in $\mathbb{Z}_p \times \mathbb{Z}_{p^n}$, then $|A| \mid p^{n+1}$.

Our proof for this direction divides two cases:

- $|A| = p$;
- $|A| = p^t, 2 \leq t \leq n$.

Tile $\Rightarrow$ Spectral: $|A| = p$

- $\mathcal{Z}_A \neq \emptyset$,
- $(x, y) \in \mathcal{Z}_A$,
- $B = \{r(x, y) : r \in [0, p-1]\}$,
- (A, B) is a spectral pair.

Tile $\Rightarrow$ Spectral: $|A| = p^t$

- (A, T) forms a tiling pair, $|A| = p^t$, $|T| = p^{n-t+1}$,
- $I = \{i \in [0, n-1] : (0, p^i) \in \mathcal{Z}_A\}$ and $J = \{i \in [0, n-1] : (0, p^i) \in \mathcal{Z}_T\}$,
- $|I| \leq t$ and $|J| \leq n - t + 1$ (by divisibility property),
- $\mathcal{Z}_A \cup \mathcal{Z}_T = \mathbb{Z}_p \times \mathbb{Z}_{p^n} \setminus \{(0, 0)\} \Rightarrow |I| + |J| \geq n$,
- $t - 1 \leq |I| \leq t$.

Tile $\Rightarrow$ Spectral: $|A| = p^t$

$$|I| = t$$

- $B = \{\sum_{i \in I} s_i(0, p^i) : s_i \in [0, p-1]\},$
- (A, B) is a spectral pair.

Tile $\Rightarrow$ Spectral: $|A| = p^t$

$|I| = t - 1$ and $|J| = n - t + 1$

- Denote $I = \{a_1, a_2, \dots, a_{t-1}\}$ and $J = \{b_1, b_2, \dots, b_{n-t+1}\}$.
- Case 1: There exist $d \in \mathbb{Z}_p$ and a_k such that $(d, p^{a_k}) \in \mathcal{Z}_T$ and $(c, p^{b_j}) \in \mathcal{Z}_T$ for all $c \in \mathbb{Z}_p$, $b_j < a_k$.
- Case 2: There exist $d \in \mathbb{Z}_p$ and b_k such that $(d, p^{b_k}) \in \mathcal{Z}_A$ and $(c, p^{a_j}) \in \mathcal{Z}_A$ for all $c \in \mathbb{Z}_p$, $a_j < b_k$.
- Case 3: $\{(c, p^{a_i}) : c \in \mathbb{Z}_p, i \in [1, t]\} \subseteq \mathcal{Z}_A$, and $\{(c, p^{b_i}) : c \in \mathbb{Z}_p, i \in [1, n - t + 1]\} \subseteq \mathcal{Z}_T$.

Tile $\Rightarrow$ Spectral: $|A| = p^t$

$$|I| = t - 1 \text{ and } |J| = n - t + 1$$

- Case 2: There exist $d \in \mathbb{Z}_p$ and b_k such that $(d, p^{b_k}) \in \mathcal{Z}_A$ and $(c, p^{a_j}) \in \mathcal{Z}_A$ for all $c \in \mathbb{Z}_p$, $a_j < b_k$.

- $B = \{s_0(d, p^{b_k}) + \sum_{i=1}^{t-1} s_i(0, p^{a_i}) : s_i \in [0, p-1]\}.$

-

$$r_0(d, p^{b_k}) + \sum_{i=1}^{t-1} r_i(0, p^{a_i}) \sim$$

$$\begin{cases} (d, p^{b_k}), & \text{if } r_0 \neq 0 \text{ and } r_i = 0 \text{ for all } a_i < b_k; \\ (r_l^{-1} r_0 d, p^{a_l}), & \text{if } r_0 \neq 0, a_l < b_k, r_l \neq 0 \text{ and } r_i = 0 \text{ for all } 1 \leq i < l; \\ (0, p^{a_l}), & \text{if } r_0 = 0, r_l \neq 0 \text{ and } r_i = 0 \text{ for all } i < l. \end{cases}$$

- (A, B) is a spectral pair.

Problem

Question

Fuglede's conjecture holds in $\mathbb{Z}_{p^m} \times \mathbb{Z}_{p^n}$?

Lemma (Equi-distributed property)

Let $A \subseteq \mathbb{Z}_{p^m} \times \mathbb{Z}_{p^n}$ and $u \in \mathbb{Z}_{p^m} \times \mathbb{Z}_{p^n}$. Then $u \in \mathcal{Z}_A$ if and only if $|H_A(u, t)| = |H_A(u, t')|$ for all $t' \equiv t \pmod{p^{n-1}}$.

Lemma (Divisibility property)

Let m, n be integers with $m \leq n$. Let $A \subseteq \mathbb{Z}_{p^m} \times \mathbb{Z}_{p^n}$. If $(a, p^{i_1}), (0, p^{i_2}), \dots, (0, p^{i_s}) \in \mathcal{Z}_A$ for some $a \in p^{m-1}\mathbb{Z}_{p^m}$ and $0 \leq i_1 < i_2 < \dots < i_s \leq n-1$, then $p^s \mid |A|$.

Cyclic groups

Fuglede's conjecture holds in

- $\mathbb{Z}_{p^n}$. (Łaba, J. Lond. Math. Soc. (2), 2002.)
- $\mathbb{Z}_{p^n q}$. (Malikiosis and Kolountzakis, Discrete Anal., 2017.)
- $\mathbb{Z}_{p^n q^2}$. (Kiss, Malikiosis, Somlai and Vizer, Anal. PDE, 2020.)
- $\mathbb{Z}_{p^m q^n}$ with $p < q$ and $m \leq 9$ or $n \leq 6$ or $p^{m-2} < q^4$. (Malikiosis, Forum Math. Sigma, 2022.)
- $\mathbb{Z}_{pqr}$. (Shi, Discrete Anal., 2019.)
- $\mathbb{Z}_{p^2 qr}$. (Somlai, Discrete Anal., 2023.)
- $\mathbb{Z}_{p^n qr}$. (Z., Combinatorica, to appear.)
- $\mathbb{Z}_{pqrs}$. (Kiss, Malikiosis, Somlai and Vizer, J. Fourier Anal. Appl., 2022.)

Cyclic groups

Tile $\rightarrow$ Spectral in $\mathbb{Z}_N$, where

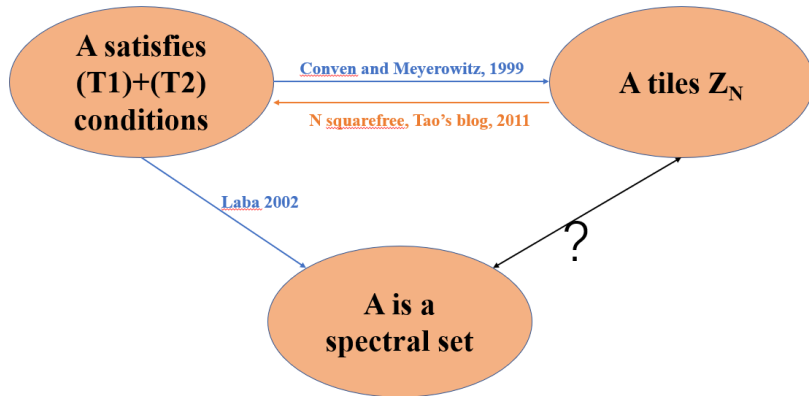
- $N = p^m q^n$. (Coven and Meyerowitz, J. Algebra, 1999)
- $N = p_1 p_2 \dots p_k$. (Tao's blog, 2011.)
- $N = p_1^n p_2 \dots p_k$. (Malikiosis, Forum Math. Sigma, 2022.)
- $N = p^2 q^2 r^2$. (Łaba and Londner, Forum Math. Pi, 2022; Invent. Math., 2023; arXiv: 2207.11809.)

method 1: (T1)+(T2) conditions

- mask polynomial: $A(X) = \sum_{a \in A} m_a X^a$, where m_a is the multiplicity of a in A .
- cyclotomic polynomial of order n : Φ_n .
- S is the set of prime powers dividing N .
- $S_A = \{s \in S : \Phi_s(X) \mid A(X)\}$.

T1,T2 conditions

- (T1) condition: $|A| = \prod_{s \in S_A} \Phi_s(1)$,
- (T2) condition: if $s_1, s_2, \dots, s_m \in S_A$ are powers of distinct primes, then $\Phi_{s_1 s_2 \dots s_m}(X) \mid A(X)$.

method 1: $(T1)+(T2)$ conditions

method 2: group ring method

The pair (A, B) being a spectral pair is equivalent to

$$|A| = |B| \text{ and } \chi_{b-b'}(A) = 0 \text{ for all } b \neq b' \in B.$$

- $\mathbb{Z}_{p^nqr} = \langle a, b, c \rangle$, where $o(a) = p^n$, $o(b) = q$ and $o(c) = r$;
- $A = \sum_{j=0}^{q-1} \sum_{k=0}^{r-1} A_{jk} b^j c^k$, where $A_{jk} \in \mathbb{Z}_{\geq 0}[\langle a \rangle]$.

Lemma (Z., in manuscript)

- (1) $p^j \in \mathcal{Z}_A$ if and only if $\chi_{p^j}(A_{jk} - A_{j0} - A_{0k} + A_{00}) = 0$ for all $j \in [0, q-1], k \in [0, r-1]$;
- (2) $p^j q \in \mathcal{Z}_A$ if and only if $\chi_{p^j}(\sum_{k=0}^{r-1} (A_{jk} - A_{j0})) = 0$ for all $k \in [0, r-1]$;
- (3) $p^j r \in \mathcal{Z}_A$ if and only if $\chi_{p^j}(\sum_{k=0}^{r-1} (A_{jk} - A_{0k})) = 0$ for all $j \in [0, q-1]$;
- (4) $p^j qr \in \mathcal{Z}_A$ if and only if $\chi_{p^j}(\sum_{j=0}^{q-1} \sum_{k=0}^{r-1} A_{jk}) = 0$.

Why so difficult?

(A, B) spectral pair $\Leftrightarrow |A| = |B|$ and $\chi_{b-b'}(A) = 0$ for all $b \neq b' \in B$.

Theorem (Lam and Leung, 2000)

Let $A \subseteq \mathbb{Z}_N$, where $N = p^a q^b$, p, q are distinct primes. If $\chi_1(A) = 0$, then $A = A_1 P + B_1 Q$, where P, Q are subgroups of $\mathbb{Z}_N$ with order p, q , respectively, and $A_1, B_1 \in \mathbb{Z}_{\geq 0}[\mathbb{Z}_N]$.

Theorem (Lam and Leung, 2000)

Let N be a positive integer and $A \subseteq \mathbb{Z}_N$. If $\chi_1(A) = 0$, then there exist integers $n_p \geq 0$ for all p prime with $p \mid N$ such that $|A| = \sum_{p \mid N} n_p p$.

Remark

Let $N = pqr$, consider $A(X) = (X^{qr} + X^{2qr} + \dots + X^{(p-1)qr})(X^{pr} + X^{2pr} + \dots + X^{(q-1)pr}) + (X^{pq} + X^{2pq} + \dots + X^{(r-1)pq})$.

Why so difficult?

A nonempty subset A of an abelian group G is defined to be **periodic** if there is an element g of G such that $A + g = A$ and $g \neq 0$.

Theorem (Szabo and Sands, Factoring groups into subsets, 2009)

If the factorization $\mathbb{Z}_n = A \oplus B$ implies that one of the subsets A, B is periodic, then n is a divisor of one of the following numbers:

$$p^n q, p^2 q^2, p^2 q r, p q r s.$$

Thank you for your attention!